

Supplementary Material

Intraoperative hypotension and hypertension are preceded by opposing autonomic signatures unified by baroreflex attenuation

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SM1. BRS Calculation Details

Supplementary Methods – BRS Calculation

Baroreflex sensitivity (BRS) was estimated per 3-minute analysis window using the sequence method. Spontaneous sequences of ≥ 3 consecutive beats were identified with concordant changes in systolic arterial pressure (SAP) and R-R interval (RRI). A sequence was accepted when the linear correlation coefficient (r) between SAP and RRI changes exceeded 0.85. BRS was computed as the average slope (ms/mmHg) of accepted sequences; windows with fewer than two accepted sequences were marked as non-calculable (NaN) and excluded from that feature's computation.

From each window, five BRS summary statistics were derived:

- brs_mean: mean slope across accepted sequences
- brs_std: standard deviation of slopes
- brs_min: minimum slope (selected for parsimonious model – most sensitive to attenuation)
- brs_max: maximum slope
- brs_slope: linear trend of BRS values over the window

BRS calculability was $\geq 96.6\%$ of windows across all cohorts and event types (see Supplementary Table 7).

SM2. Lead-time Back-extrapolation

Supplementary Methods – Lead-time Back-extrapolation

The reference window (lead time = 0) is the 3-minute window immediately preceding event onset. Lead-time analysis extrapolates AUC as a function of advance warning. For each lead time $L \in \{5, 10, 15, 20, 30\}$ min:

1. The GLMM linear predictor is computed for the reference window ($L=0$).
2. The predictor is linearly attenuated by a factor $(1 - L/TMAX)$, where $TMAX = 45$ min (conservative estimate of signal degradation).
3. AUC is re-computed from the attenuated predictor using out-of-sample cross-validation (leave-one-patient-out on development cohort).

This approach provides a conservative lower bound on discriminative ability at advance lead times. Results are reported for the Clínic development cohort; external validation on VitalDB is ongoing.

SM3. MILP Optimisation

Supplementary Methods – MILP Optimisation (SM3)

The MILP (Mixed-Integer Linear Programming) depth-2 tree was optimised to maximise sensitivity at a specified specificity target ($\geq 80\%$) on the Clinic development set, subject to interpretability constraints (depth ≤ 2 splits, binary threshold decisions).

Optimisation was performed using the Python PuLP library (v2.x) with the CBC MILP solver. A bootstrap procedure ($B = 500$ replicates, stratified by patient) was used to estimate the sampling variability of rule thresholds. Only thresholds present in $>50\%$ of bootstrap fits were retained in the final rule (threshold stability). The final rules use the median threshold across stable bootstrap replicates.

The MILP is intended as a tool for the translation of the autonomic signature into an actionable clinical rule at the bedside. It is NOT the primary predictive model – the GLMM is the primary model for AUC computation.

Supplementary Table 1. Univariate AUC for all 40 signal features across both cohorts and both outcomes. AUC computed from parquet windows; AUC < 0.5 flipped (feature used in inverted direction).

Feature	Domain	AUC Clínic Hypo	AUC VitalDB Hypo	AUC Clínic Hyper	AUC VitalDB Hyper
ARV max	BPV	0.593	0.774	0.649	0.714
ARV mean	BPV	0.548	0.531	0.62	0.575
ARV min	BPV	0.587	0.677	0.511	0.572
ARV slope	BPV	0.531	0.567	0.561	0.578
ARV std	BPV	0.617	0.781	0.68	0.724
CV(PA) max	BPV	0.551	0.704	0.75	0.688
CV(PA) mean	BPV	0.614	0.603	0.759	0.647
CV(PA) min	BPV	0.702	0.639	0.58	0.527
CV(PA) slope	BPV	0.59	0.715	0.511	0.654
CV(PA) std	BPV	0.639	0.711	0.702	0.652
SD(PA) max	BPV	0.728	0.718	0.666	0.663
SD(PA) mean	BPV	0.575	0.654	0.664	0.65
SD(PA) min	BPV	0.569	0.596	0.514	0.532
SD(PA) slope	BPV	0.626	0.76	0.556	0.675
SD(PA) std	BPV	0.75	0.721	0.67	0.652
BRS max	BRS	0.518	0.503	0.512	0.581
BRS mean	BRS	0.54	0.528	0.516	0.597
BRS minimum	BRS	0.6	0.601	0.611	0.575
BRS slope	BRS	0.535	0.535	0.531	0.524
BRS std	BRS	0.504	0.519	0.501	0.582
RMSSD max	HRV	0.553	0.633	0.514	0.684
RMSSD mean	HRV	0.536	0.604	0.548	0.634
RMSSD min	HRV	0.544	0.523	0.593	0.502
RMSSD slope	HRV	0.541	0.608	0.515	0.632
RMSSD std	HRV	0.548	0.592	0.501	0.639
SDNN max	HRV	0.502	0.631	0.523	0.674
SDNN mean	HRV	0.584	0.614	0.544	0.644
SDNN min	HRV	0.582	0.533	0.607	0.507
SDNN slope	HRV	0.555	0.604	0.507	0.639
SDNN std	HRV	0.512	0.589	0.504	0.635
pNN50 max	HRV	0.554	0.719	0.55	0.758
pNN50 mean	HRV	0.582	0.628	0.528	0.646
pNN50 min	HRV	0.525	0.52	0.525	0.513
pNN50 slope	HRV	0.521	0.589	0.52	0.636
pNN50 std	HRV	0.587	0.638	0.534	0.657
RSA max	RSA	0.517	0.559	0.586	0.606
RSA mean	RSA	0.621	0.58	0.573	0.612
RSA min	RSA	0.591	0.505	0.551	0.509
RSA slope	RSA	0.524	0.562	0.525	0.606
RSA std	RSA	0.502	0.537	0.548	0.599

Supplementary Table 2. Bootstrap sign stability for parsimonious GLMM features (B=1,000). Features with stability <75% should not be interpreted physiologically.

Feature	Domain	Outcome	Point estimate	Dominant sign	Sign stability (%)	Interpretation
BRS minimum	BRS	Hypertension	-1.5242	–	100.0	directionally robust
CV(PA) std	BPV	Hypertension	-2.5405	–	99.9	directionally robust
ARV std	BPV	Hypertension	0.9729	+	99.9	directionally robust
CV(PA) mean	BPV	Hypertension	-1.3342	–	99.7	directionally robust
SD(PA) std	BPV	Hypertension	3.3961	+	99.1	directionally robust
SD(PA) slope	BPV	Hypertension	0.5759	+	93.3	directionally robust
SD(PA) max	BPV	Hypertension	-1.1254	–	90.1	directionally robust
SDNN mean	HRV	Hypertension	-0.1124	–	64.3	do not interpret physiologically
BRS minimum	BRS	Hypotension	-1.7004	–	100.0	directionally robust
CV(PA) std	BPV	Hypotension	1.3287	+	99.2	directionally robust
SD(PA) max	BPV	Hypotension	-0.7667	–	99.2	directionally robust
SD(PA) mean	BPV	Hypotension	0.9001	+	96.5	directionally robust
RSA max	RSA	Hypotension	-0.5653	–	89.2	directionally moderate
RSA mean	RSA	Hypotension	0.0524	+	59.2	do not interpret physiologically
ARV std	BPV	Hypotension	-0.0238	–	58.3	do not interpret physiologically
ARV mean	BPV	Hypotension	0.0025	+	52.7	do not interpret physiologically

Supplementary Table 3. VitalDB cohort demographic characteristics.

Variable	All included (n=1,080)	Source
Age, years [median (IQR)]	61 (51–69)	VitalDB cases.csv
Female sex (%)	41.7%	VitalDB cases.csv
BMI, kg/m ² [median (IQR)]	22.6 (20.4–25.0)	VitalDB cases.csv
ASA class III–IV (%)	18.9%	VitalDB cases.csv
Emergency surgery (%)	13.7%	VitalDB cases.csv
Recording duration, h [median]	3.08	Derived
Hypotension events (n patients)	505 (580)	BeatLabile pipeline
Hypertension events (n patients)	75 (306)	BeatLabile pipeline
Total windows	4,471	BeatLabile pipeline

Supplementary Table 4. Pharmacological stratification results on VitalDB hold-out (30%, n=324 patients). M1 = drug-naïve signal model; M2 = full signal model.

Outcome	Stratum	N events	N controls	AUC M1 (drug-naïve signal)	AUC M1 95% CI	AUC M2 (signal)
Hypotension	All	505	483	0.58	[0.541-0.616]	0.75
Hypotension	Any Vaso	407	389	0.566	[0.524-0.607]	0.688
Hypotension	No Vaso	98	94	0.643	[0.556-0.726]	0.874
Hypotension	Stratum Eph Only	204	196	0.559	[0.503-0.622]	0.854
Hypotension	Stratum Phe Epi Bolus	123	124	0.588	[0.508-0.661]	0.724
Hypotension	Stratum No Vaso	98	94	0.643	[0.556-0.726]	0.874
Hypertension	All	75	483	0.56	[0.479-0.642]	0.817
Hypertension	Any Vaso	55	389	0.552	[0.469-0.647]	0.821

Supplementary Table 5. Lead-time analysis (Clínic). AUC estimated by linear attenuation of GLMM predictor at successive advance times.

Outcome	Lead time (min)	AUC	95% CI	N events
Hypotension	0	0.7839	[0.678-0.855]	66
Hypotension	5	0.6685	[0.505-0.783]	66
Hypotension	10	0.6246	[0.462-0.758]	66
Hypotension	15	0.605	[0.436-0.746]	66
Hypotension	20	0.5927	[0.420-0.737]	66
Hypotension	30	0.5677	[0.384-0.721]	66
Hypertension	0	0.8989	[0.791-0.966]	38
Hypertension	5	0.8527	[0.704-0.941]	38
Hypertension	10	0.8045	[0.623-0.915]	38
Hypertension	15	0.7794	[0.570-0.902]	38
Hypertension	20	0.7657	[0.555-0.894]	38
Hypertension	30	0.7491	[0.535-0.885]	38

Supplementary Table 6. Duration sensitivity analysis. Events re-classified using progressively stricter duration thresholds.

Outcome	Duration threshold	N events	Prevalence	GLMM AUC
Hypotension	1min	66	0.0694	nan
Hypotension	3min	66	0.0694	nan
Hypotension	5min	28	0.0294	nan
Hypotension	10min	8	0.0084	nan
Hypertension	1min	38	0.0412	nan
Hypertension	3min	38	0.0412	nan
Hypertension	5min	14	0.0152	nan
Hypertension	10min	4	nan	nan

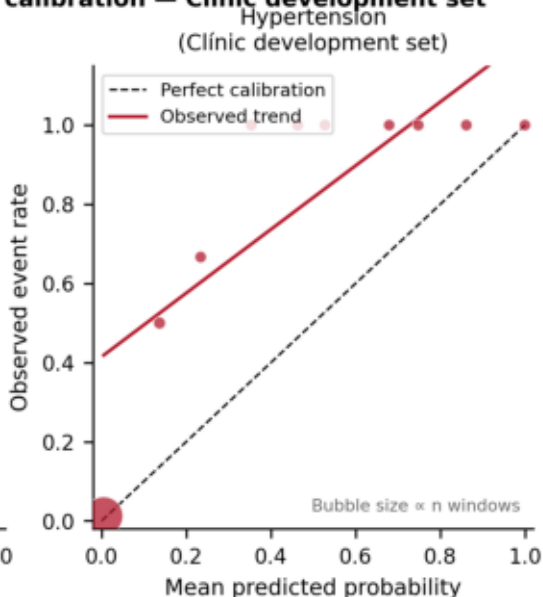
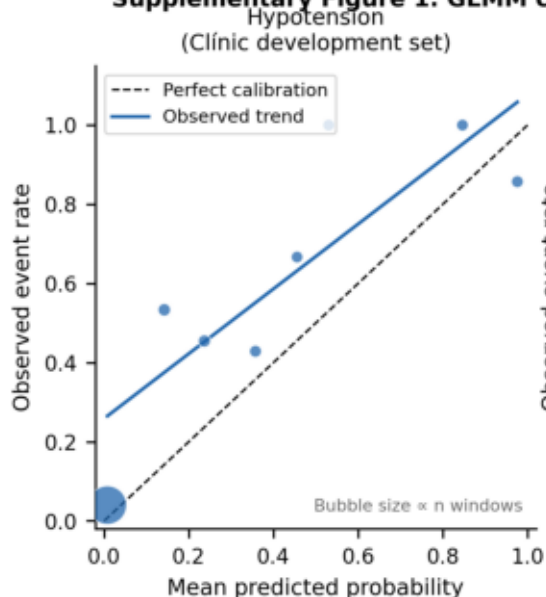
Supplementary Table 7. BRS feature calculability across cohorts and event types. BRS calculable = ≥ 2 valid sequences in window.

Cohort	Outcome	N windows (total)	BRS calculable (%)	BRS NaN (%)
Clínic	Hypotension	951	96.64	3.36
Clínic	Variability	2481	94.6	5.4
Clínic	Hypertension	923	96.97	3.03
VitalDB	Variability	2925	96.41	3.59
VitalDB	Hypotension	988	95.85	4.15
VitalDB	Hypertension	558	97.31	2.69

Supplementary Table 8. MILP depth-2 decision rule operating characteristics on development (Clínic) and test (VitalDB) sets.

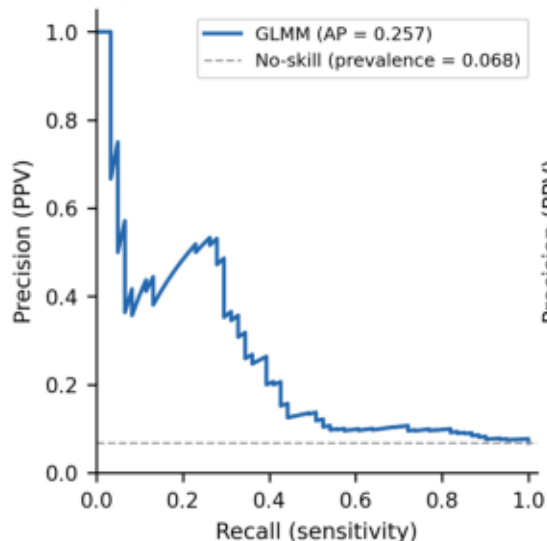
Outcome	Dataset	N windows	Sensitivity	Specificity	PPV	NPV	F1	LR+	LR−
Hypotension	Clinic Development	951	0.485	0.823	0.169	0.955	0.251	2.73	0.63
Hypotension	Vitaldb Test	287	0.513	0.77	0.716	0.584	0.598	2.23	0.63
Hypotension	Vitaldb Full	988	0.511	0.783	0.711	0.605	0.595	2.35	0.62
Hypertension	Clinic Development	923	0.605	0.858	0.154	0.981	0.246	4.25	0.46
Hypertension	Vitaldb Test	167	0.087	0.965	0.286	0.869	0.133	2.5	0.95
Hypertension	Vitaldb Full	558	0.12	0.932	0.214	0.872	0.154	1.76	0.94

Supplementary Figure 1. GLMM calibration — Clínic development set

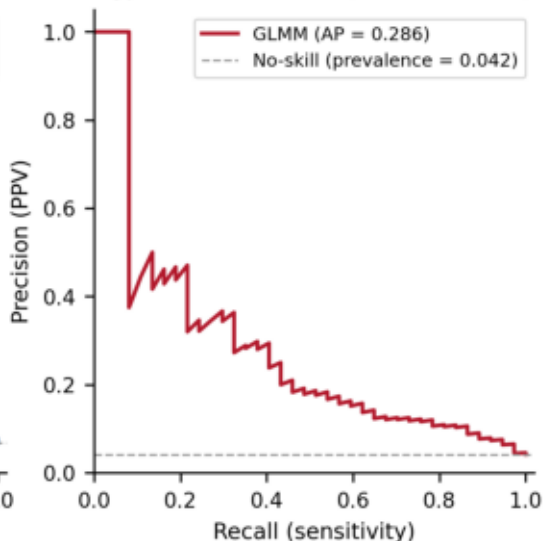


Supplementary Figure 2. Precision-recall curves — parsimonious GLMM

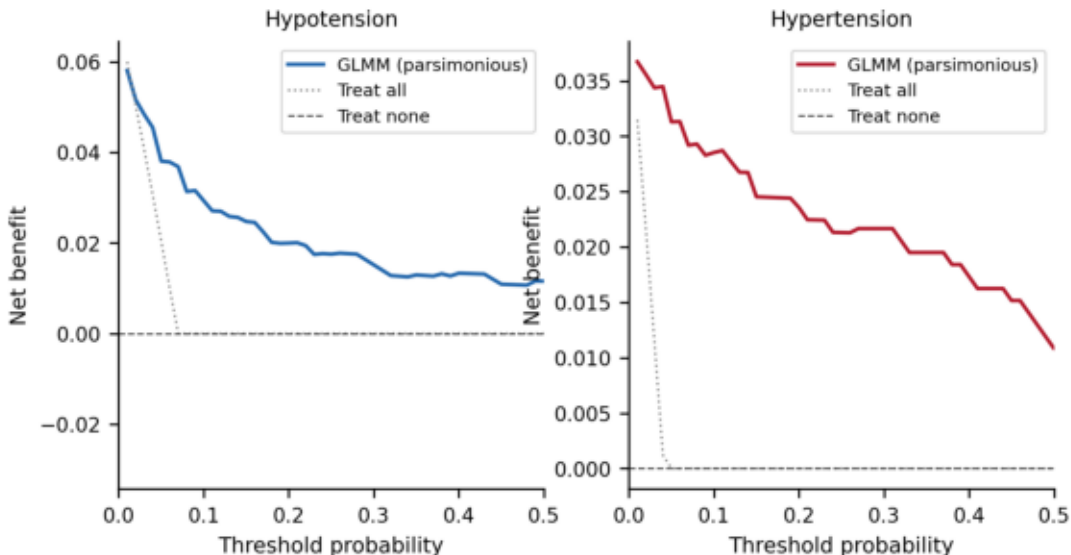
Hypotension — Clínic (cross-validated)



Hypertension — Clínic (cross-validated)



Supplementary Figure 3. Decision curve analysis — Clínic development set



Translational implications: MILP decision rules

MILP-derived decision rules (depth-2, B=500 bootstrap replicates)

Hypotension:

Alert if ($CV(PA)_{-min} \leq 0.0225$ AND $CV(PA)_{-mean} > 0.0547$)
OR ($CV(PA)_{-min} > 0.0225$ AND $SD(PA)_{-mean} > 8.05$ mmHg)

Hypertension:

Alert if ($CV(PA)_{-mean} \leq 0.0525$ AND $SD(PA)_{-max} > 8.70$ mmHg)
OR ($CV(PA)_{-mean} > 0.0525$ AND $SD(PA)_{-max} \leq 3.94$ mmHg)

Interpretation:

The MILP rules translate the continuous autonomic signature into binary bedside-applicable alerts using two features: CV(PA) and SD(PA). These are computable in real-time from any arterial waveform monitor.

Importantly:

- The hypotension rule achieves clinically useful NPV (Clínica: 0.956; VitalDB: 0.584) at the cost of lower sensitivity (~50%), consistent with the prediction 10–30 min before MAP < 65 mmHg.
- The hypertension rule transfers poorly to VitalDB (sensitivity 8.7%), likely reflecting differences in BP management practice between settings. This limitation warrants prospective validation before clinical deployment.

The MILP rules are provided as illustrative examples of knowledge translation. They are NOT intended for clinical use without prospective validation.

STROBE Checklist for Observational Studies

Item	Section/topic	Recommendation (abbreviated)	Manuscript location
1a	Title	Indicates observational study design	Title, Methods
1b	Abstract	Informative abstract	Abstract
2	Background/rationale	Scientific background and rationale	Introduction
3	Objectives	Specific objectives	Introduction
4	Study design	Presents key elements of study design	Methods - Study design
5	Setting	Describes setting, locations, relevant dates	Methods - Setting
6	Participants	Eligibility criteria, source/methods of selection	Methods - Participants & Figure 1
7	Variables	Defines all outcomes, exposures, predictors	Methods - Variables
8	Data sources/measurement	For each variable, data sources and assessment methods	Methods - Signal processing
9	Bias	Describes efforts to address potential sources of bias	Methods - Signal quality QC
10	Study size	Explains how study size was arrived at	Methods - Sample size / ST3
11	Quantitative variables	Explains how quantitative variables were handled	Methods - Feature extraction
12a	Statistical methods	All statistical methods, including for confounding	Methods - Statistical analysis
12b	Statistical - subgroups	Methods for any subgroup analyses	Methods - Sensitivity analyses / ST4
12c	Statistical - missing data	How missing data were addressed	Methods - QC pipeline / ST7
12e	Statistical - sensitivity	Any sensitivity analyses	Methods - Sensitivity analyses
13a	Participants flow	Numbers of eligible, excluded, included	Figure 1 (STROBE)
13b	Non-participation	Reasons for non-participation at each stage	Figure 1
13c	Flow diagram	Consider use of a flow diagram	Figure 1
14a	Descriptive data	Characteristics of study participants	ST3 / Results - Cohorts
14c	Missing data	Number of missing values for each variable of interest	ST7 / Methods
15	Outcome data	Report numbers of outcome events	Results, ST3
16a	Main results	Give unadjusted and adjusted estimates (if applicable)	Results - GLMM
16b	Precision	Report category boundaries for continuous variables	Methods
16c	If ≥10 outcome events	Consider translating into absolute risk for a meaningful tim	Results - lead-time analysis
17	Other analyses	Report other analyses performed	Supplementary / sensitivity
18	Key results	Summarises key results with reference to study objectives	Discussion
19	Limitations	Discusses limitations, sources of potential bias	Discussion - Limitations
20	Interpretation	Gives overall interpretation, considering evidence	Discussion
21	Generalisability	Discusses generalisability of study results	Discussion - Generalisability
22	Funding	Gives funding source and role of funders	Funding statement